Introduction to R Programming

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1 Introduction

R is a programming language and environment widely used for: - Statistical analysis - Data visualization - Data science and machine learning - Academic and industry research

This document introduces the **fundamental concepts of R** using simple and reproducible examples.

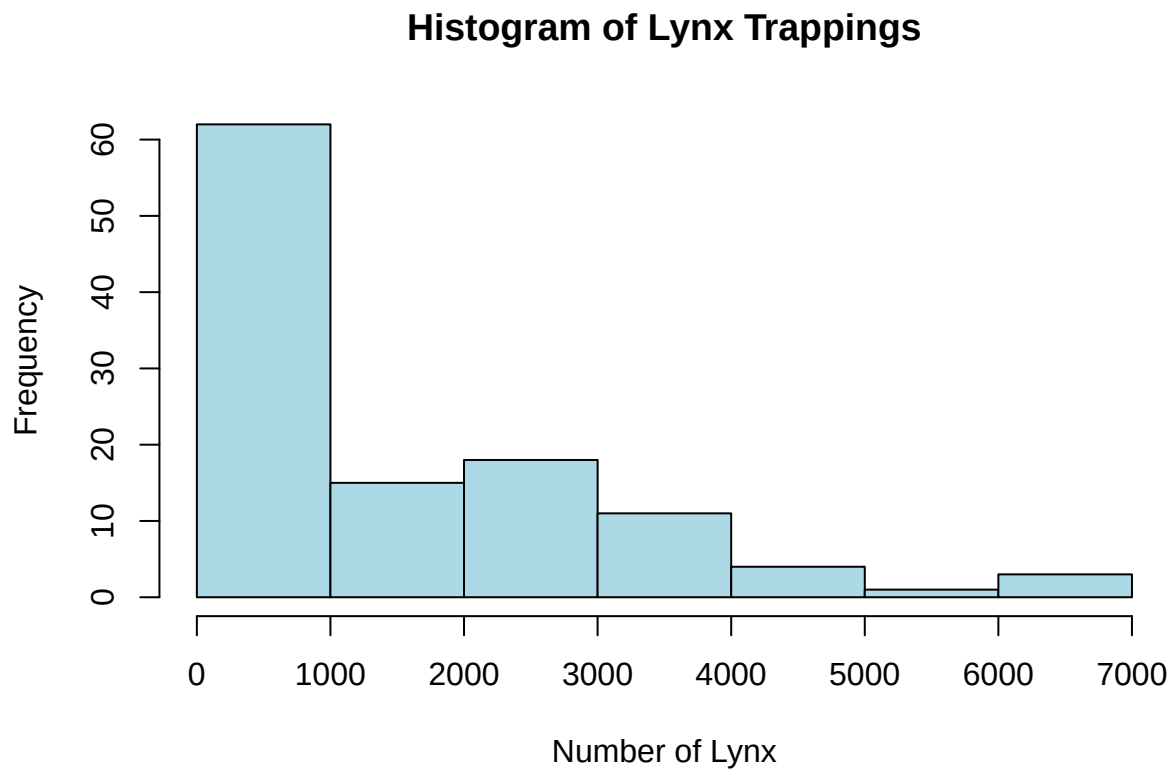
---

2 Using Built-in Data

R includes several built-in datasets for learning and experimentation.

The `lynx` dataset records annual numbers of lynx trappings in Canada.

```
hist(lynx,
     main = "Histogram of Lynx Trappings",
     col = "lightblue",
     xlab = "Number of Lynx")
```



Key idea:
A histogram is used to understand the distribution of a numeric variable.

---

3 Basic Arithmetic in R

R supports all standard arithmetic operations.

```
12 + 3          # Addition
```

```
## [1] 15
```

```
(12 / 3) * 3    # Division and multiplication
```

```
## [1] 12
```

```
(45 / 5) * 12 + 10
```

```
## [1] 118
```

```r
(23 * 3) - (43 * 2)
```

```
## [1] -17
```

3.1 Assignment Operator

Values are stored in variables using the assignment operator `<-`.

```r
X <- (45 / 5) * 12 + 10
Y <- (23 * 3) - (43 * 2)

X - Y
```

```
## [1] 135
```

Note:
Although `=` can be used, `<-` is the conventional assignment operator in R.

---

4 Getting Help and Documentation in R

R provides extensive **built-in documentation** for all functions, datasets, and packages. This documentation can be accessed directly from **RStudio**.

4.1 Using the `?` Operator

The question mark `?` is a shortcut to access help files.

```r
?hist
```

This opens the help page for the `hist()` function in the **Help** pane of RStudio.

4.2 Using the `help()` Function

The `help()` function performs the same task as the `?` operator.

```r
help(mean)
```

4.3 Getting Help for Datasets

You can also view documentation for built-in datasets.

```
?iris
```

4.4 Searching for Help When You Are Unsure

If you are not sure about the exact function name, use `help.search()`.

```
help.search("regression")
```

4.5 Why This Is Important

Reading the documentation helps you understand: - What a function does - What arguments it takes - What values it returns - Examples of correct usage

Learning to use R's help system is a key step toward becoming independent in R programming.

5 Variables and Vectors

A **vector** is the most basic data structure in R.
All elements in a vector must be of the same data type.

```
PARatings <- c(3, 2, 4, 3, 5, 4, 3, 2, 3, 4, 5)

Name <- c("Ravi", "Sudha", "Ramesh", "Soham", "Shilpa",
          "Bhavesh", "Amarpreet", "Wilfred", "Abdul", "Srini", "Taposh")

Age <- c(23, 22, 23, 25, 26, 24, 27, 23, 23, 25, 26)
```

---

6 Data Frames

A **data frame** is a tabular data structure where: - Each column is a vector - Each row is an observation

```
PerfData <- data.frame(Name, PARatings, Age)
PerfData
```

```
##         Name PARatings Age
## 1       Ravi         3  23
## 2      Sudha         2  22
## 3     Ramesh         4  23
```

```
## 4      Soham         3  25
## 5     Shilpa         5  26
## 6    Bhavesh         4  24
## 7  Amarpreet         3  27
## 8    Wilfred         2  23
## 9      Abdul         3  23
## 10     Srini         4  25
## 11    Taposh         5  26
```

6.1 Inspecting a Data Frame

```r
dim(PerfData)      # Dimensions (rows, columns)
```

```
## [1] 11  3
```

```r
str(PerfData)      # Structure of the data
```

```
## 'data.frame':    11 obs. of  3 variables:
##  $ Name     : chr  "Ravi" "Sudha" "Ramesh" "Soham" ...
##  $ PARatings: num  3 2 4 3 5 4 3 2 3 4 ...
##  $ Age      : num  23 22 23 25 26 24 27 23 23 25 ...
```

```r
summary(PerfData)  # Summary statistics
```

```
##      Name             PARatings          Age
##  Length:11          Min.   :2.000   Min.   :22.00
##  Class :character   1st Qu.:3.000   1st Qu.:23.00
##  Mode  :character   Median :3.000   Median :24.00
##                     Mean   :3.455   Mean   :24.27
##                     3rd Qu.:4.000   3rd Qu.:25.50
##                     Max.   :5.000   Max.   :27.00
```

These functions are essential for initial data exploration.

---

7 Working with Files (Reproducible Approach)

Avoid using `setwd()` as it makes code non-reproducible.
Instead, read and write files relative to the project directory.

```r
write.csv(PerfData, "Appraisal.csv", row.names = FALSE)
PerfData1 <- read.csv("Appraisal.csv")
```

```r
str(PerfData1)
```

```
## 'data.frame':    11 obs. of  3 variables:
##  $ Name     : chr  "Ravi" "Sudha" "Ramesh" "Soham" ...
##  $ PARatings: int  3 2 4 3 5 4 3 2 3 4 ...
##  $ Age      : int  23 22 23 25 26 24 27 23 23 25 ...
```

```r
summary(PerfData1)
```

```
##      Name             PARatings          Age
##  Length:11          Min.   :2.000   Min.   :22.00
##  Class :character   1st Qu.:3.000   1st Qu.:23.00
##  Mode  :character   Median :3.000   Median :24.00
##                     Mean   :3.455   Mean   :24.27
##                     3rd Qu.:4.000   3rd Qu.:25.50
##                     Max.   :5.000   Max.   :27.00
```

---

8 Built-in Dataset: Iris

The `iris` dataset is commonly used for demonstrating classification and visualization techniques.

```r
data(iris)
str(iris)
```

```
## 'data.frame':    150 obs. of  5 variables:
##  $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
##  $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
##  $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
##  $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
##  $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
```

```r
summary(iris)
```

```
##   Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
##  Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
##  1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
##  Median :5.800   Median :3.000   Median :4.350   Median :1.300
##  Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
##  3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
##  Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
##        Species
##  setosa    :50
##  versicolor:50
##  virginica :50
##
##
##
```

```r
head(iris)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1         3.5          1.4         0.2  setosa
## 2          4.9         3.0          1.4         0.2  setosa
## 3          4.7         3.2          1.3         0.2  setosa
## 4          4.6         3.1          1.5         0.2  setosa
## 5          5.0         3.6          1.4         0.2  setosa
## 6          5.4         3.9          1.7         0.4  setosa
```

```r
tail(iris)
```

```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
## 145          6.7         3.3          5.7         2.5 virginica
## 146          6.7         3.0          5.2         2.3 virginica
## 147          6.3         2.5          5.0         1.9 virginica
## 148          6.5         3.0          5.2         2.0 virginica
## 149          6.2         3.4          5.4         2.3 virginica
## 150          5.9         3.0          5.1         1.8 virginica
```

---

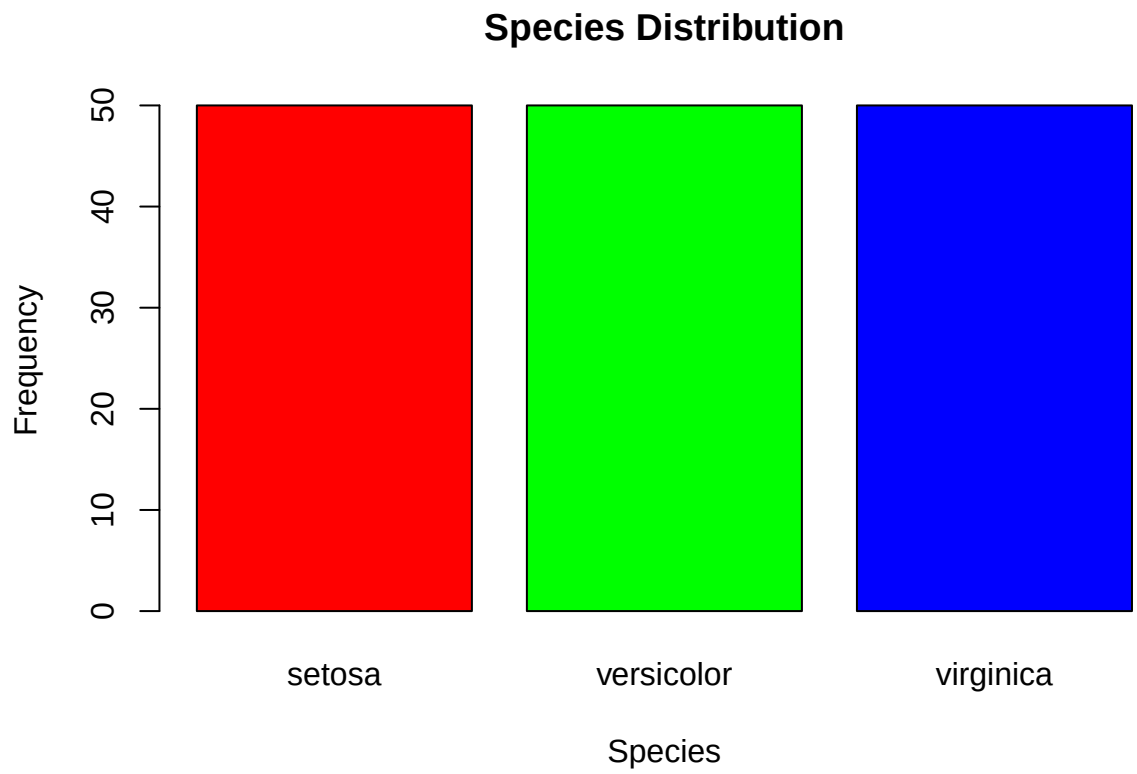
9 Visualizing Categorical Data

Categorical (qualitative) data represents groups or categories.

9.1 Bar Plot

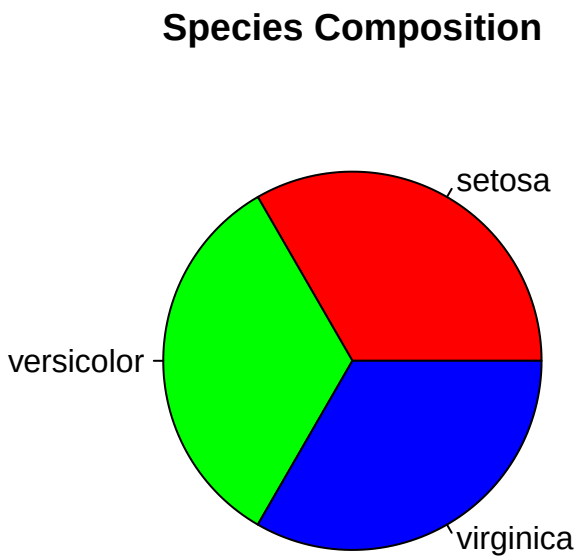
```r
tab <- table(iris$Species)

barplot(tab,
        main = "Species Distribution",
        xlab = "Species",
        ylab = "Frequency",
        col = rainbow(3))
```



9.2 Pie Chart

```
pie(tab,
    main = "Species Composition",
    col = rainbow(3))
```



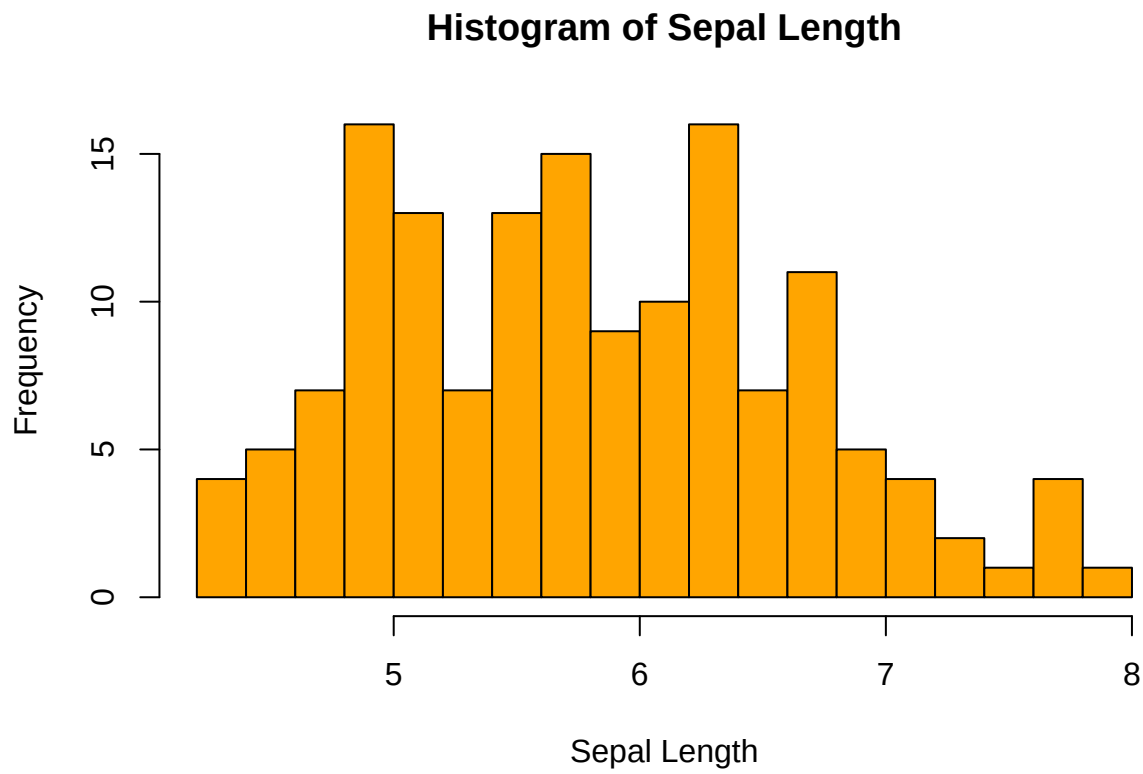
Bar plots and pie charts are commonly used to show category frequencies.

---

10 Visualizing Numerical Data

10.1 Histogram

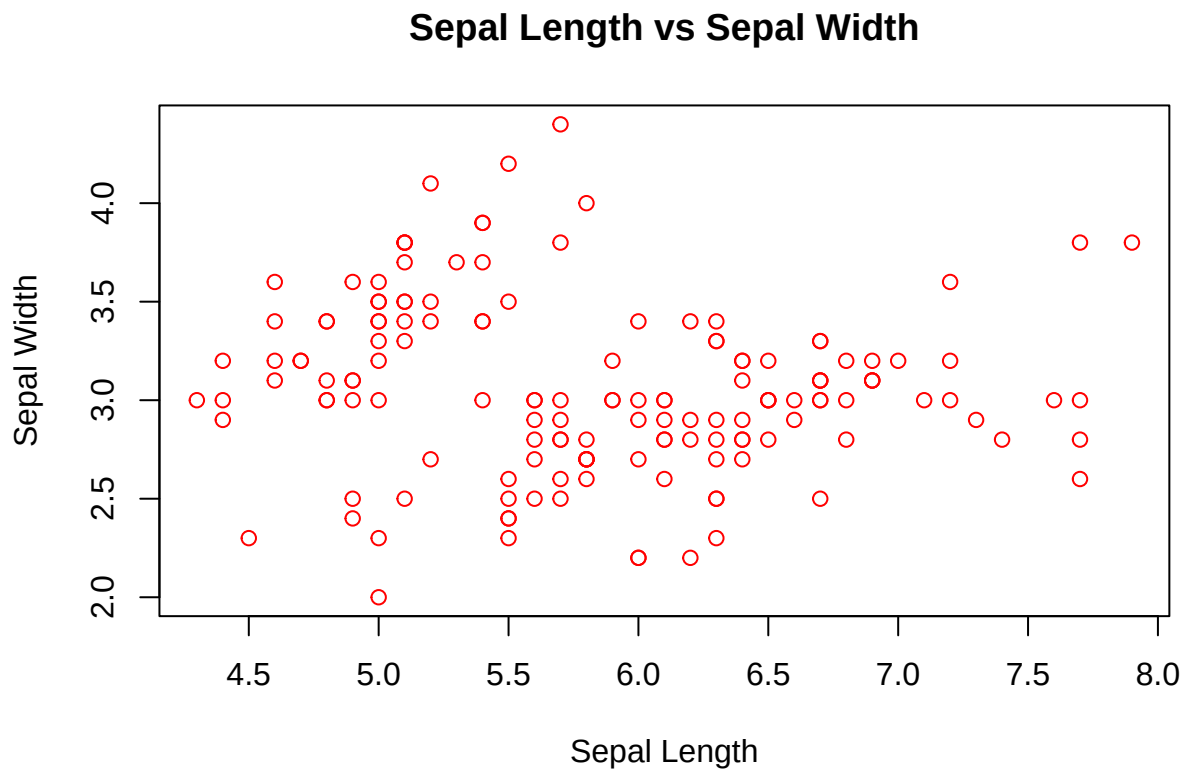
```
hist(iris$Sepal.Length,
     main = "Histogram of Sepal Length",
     xlab = "Sepal Length",
     col = "orange",
     breaks = 15)
```



Histograms show the shape, center, and spread of numeric data.

10.2 Scatter Plot

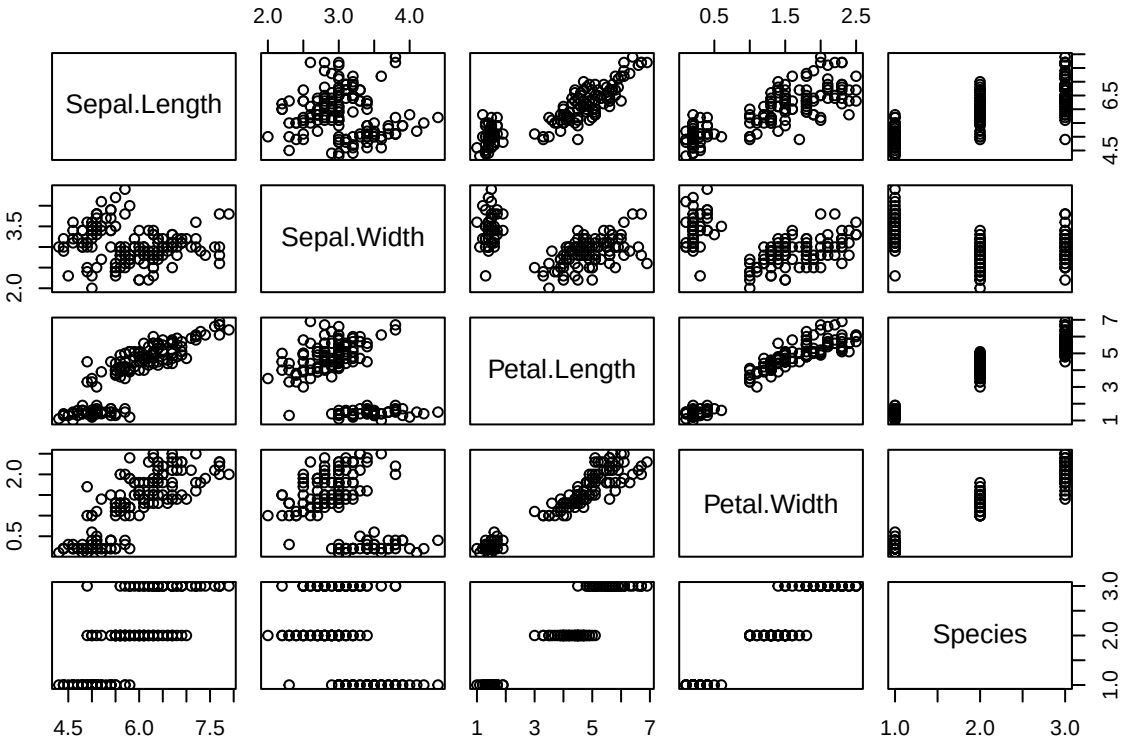
```
plot(iris$Sepal.Length, iris$Sepal.Width,
     col = "red",
     main = "Sepal Length vs Sepal Width",
     xlab = "Sepal Length",
     ylab = "Sepal Width")
```



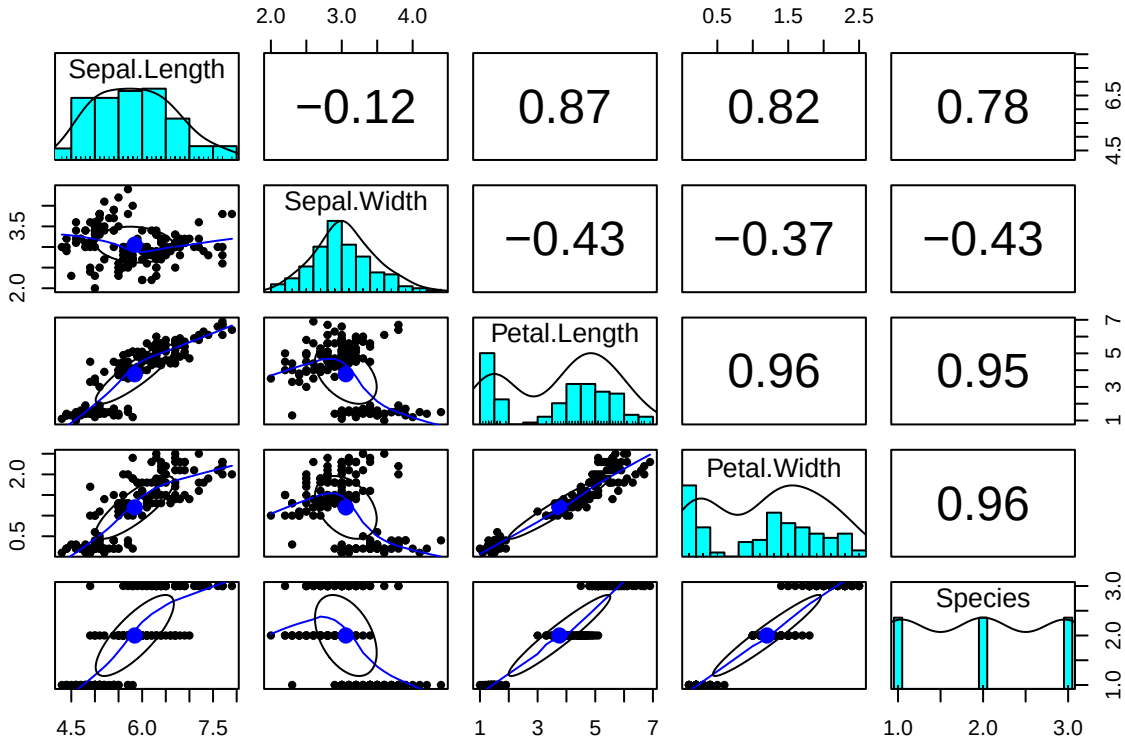
Scatter plots visualize relationships between two numeric variables.

10.3 Multiple Scatter Plots and Correlations

```
pairs(iris)
```




```r
library(psych)
pairs.panels(iris)
```



These plots help identify correlations among multiple variables.

---

11 Data Types in R

R commonly works with numeric, character, and logical data types.

```
A <- 4
class(A)
```

```
## [1] "numeric"
```

```
B <- "gender"
class(B)
```

```
## [1] "character"
```

```
C <- TRUE
class(C)
```

```
## [1] "logical"
```

---

12 Vectors, Coercion, and Lists

12.1 Named Vectors

```
Age <- c(Ravi = 23, Sudha = 22, Ramesh = 23, Soham = 25, Shilpa = 26)
Age
```

```
##   Ravi  Sudha Ramesh  Soham Shilpa
##     23     22     23     25     26
```

12.2 Type Coercion

When different data types are combined in a vector, R coerces them to a common type.

```
mixed <- c(23, 22, "Ravi", 25)
class(mixed)
```

```
## [1] "character"
```

Character data has higher priority than numeric data.

12.3 Lists

Lists can store elements of different data types.

```
mylist <- list(23, 22, "Ravi", TRUE)
sapply(mylist, class)
```

```
## [1] "numeric"   "numeric"   "character" "logical"
```

---

13 Factors (Categorical Variables)

Factors are used to represent categorical and ordinal data.

```
PARatings <- c(3, 2, 4, 3, 5, 4, 3, 2, 3, 4, 5)

PARatings_factor <- factor(PARatings)

PARatings_ordered <- factor(PARatings,
                            ordered = TRUE,
                            levels = c(2, 3, 4, 5))

str(PARatings_ordered)
```

```
##  Ord.factor w/ 4 levels "2"<"3"<"4"<"5": 2 1 3 2 4 3 2 1 2 3 ...
```

Ordered factors are useful when category levels have a natural order.

---

14 Conditional Statements

Conditional statements allow R to make decisions based on logical conditions.

```r
x <- 10

if (x > 0) {
  "x is positive"
} else {
  "x is negative"
}
```

```
## [1] "x is positive"
```

```r
if (iris$Sepal.Length[1] < 4) {
  "Sepal length is less than 4"
} else {
  "Sepal length is greater than or equal to 4"
}
```

```
## [1] "Sepal length is greater than or equal to 4"
```

---

15 Operators in R

15.1 Arithmetic Operators

```r
A <- 12
B <- 10

A * B
```

```
## [1] 120
```

```r
A - B
```

```
## [1] 2
```

15.2 Relational Operators

```r
A < B
```

```
## [1] FALSE
```

```r
A > B
```

```
## [1] TRUE
```

```r
A == B
```

```
## [1] FALSE
```

```r
A != B
```

```
## [1] TRUE
```

15.3 Logical Operators

```r
p <- 10

(p < 15 & p > 8)    # AND
```

```
## [1] TRUE
```

```r
(p > 15 | p > 8)    # OR
```

```
## [1] TRUE
```

```r
!(p < 15)           # NOT
```

```
## [1] FALSE
```

---

16 Conclusion

This document introduced the foundational concepts of R programming, including: - Data structures - Visualization techniques - File handling - Conditional logic and operators

These fundamentals form the basis for advanced topics such as statistics, data analysis, and machine learning.